

YASH AGNIHOTRI

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[Linkedin](#) | [Github](#) | [Portfolio](#) | [Leetcode](#)

PROFILE

Full-stack engineer and ML enthusiast with hands-on experience building scalable backends, AI-driven systems, and hardware-software integration. Strong foundation in data structures, algorithms, and system design. Seeking roles in backend engineering, AI/ML, or full-stack development.

EDUCATION

Vellore Institute of Technology-AP — B.Tech, Computer Science Engineering	Sep 2023 – Present
CGPA: 9.01 / 10.0 Amaravati, Andhra Pradesh	
Kendriya Vidyalaya, Hardoi — CBSE Class XII (94.6%)	Apr 2021 – Apr 2022
Kendriya Vidyalaya, Hardoi — CBSE Class X (88.4%)	Mar 2019 – Jun 2020

EXPERIENCE

SmartBridge — Generative AI Summer Intern	Jun 2025 – Jul 2025
- Architected Mistral-based dialogue-generation framework into a functional AI game-dialogue generator: rebuilt the backend in Flask and added a Streamlit interface for real-time interaction. - Redesigned context-injection logic in the Mistral pipeline to preserve NPC persona and memory across multi-turn exchanges, eliminating context drift observed in the baseline. - Reduced LLM response generation latency to <1.8 seconds per query by optimizing prompt structure and token constraints. Optimized NPC context memory window, reducing average prompt size from 1,000 to 800 tokens (20%) while maintaining multi-turn conversational coherence.	

PROJECTS

Career Compass AI	Python, Flask, Node Js, XG Boost
- Benchmarked 5 regression architectures (XGBoost, Random Forest, Gradient Boosting, Decision Tree, Linear Regression) to predict candidate career-fit/compensation from profile features - best model (XGBoost) reached $R^2 = 0.79$, $MAE \approx \\$46,833$, $RMSE \approx \\$65,913$ in 1.08s training time, revealing feature-engineering gaps limiting further gains	
AI Crop Yield Prediction & Optimization	Python, TensorFlow, Neural Networks
- Achieved a 0.88 R^2 score across multi-variable agricultural datasets, mitigating overfitting via Dropout layers and L2 regularization. - Optimized the TensorFlow inference pipeline to process soil and weather data inputs in under 35 milliseconds.	
Music-Recommendier	Keras, TensorFlow, Spotipy, Tkinter, Flask
- Developed a real-time emotion-aware music recommendation system using CNN-based facial expression recognition trained on the FER-2013 dataset, classifying 7 emotions with ~72% validation accuracy. - Built a multithreaded Flask application achieving 25–30 FPS real-time webcam inference with optimized video streaming and preprocessing. - Designed an end-to-end inference pipeline using OpenCV, face detection, and TensorFlow, processing 48×48 grayscale images with an average prediction time of <40 ms per frame.	

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL

Web & Backend: HTML, CSS, React, Node.js, Express.js, Flask, Streamlit, REST APIs

Databases: MySQL, MongoDB

ML/AI & Data: NumPy, Pandas, Scikit-learn, XGBoost, OpenCV, Matplotlib, Generative AI/LLMs (Mistral)

Cloud & DevOps: AWS (EC2/S3), Kubernetes, RPA, Docker

Tools: Git, GitHub, VS Code, Google Colab, Postman

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

CODING PROFILE

- LeetCode: 400+ problems solved (Java-focused). Strongest tags: DP, Backtracking, Divide & Conquer, Hash Table.

CERTIFICATIONS

AWS Certified Cloud Practitioner (AWS) | IBM Cloud Certification (IBM) | Generative AI (GenAI) Course (SmartBridge) | Web 3.0 Certification (Teachnook)